AI Playbook and Toolkit for Public Health Departments

FHIR for Public Health AI Use Cases

ARC 250

FHIR is increasingly important for public health because it supports structured, API-based access to health data. Public health staff do not need to become FHIR developers to participate in AI planning, but they do need to understand what FHIR can and cannot provide. This module explains how FHIR resources, profiles, implementation guides, and APIs may support AI-enabled public health workflows.

Level	applied/foundational
Primary track	Technical Architecture Track
Appears in tracks	technical-architecture

Learning Objectives

- Define FHIR in practical public health terms.
- Identify FHIR resources commonly relevant to public health AI use cases.
- Explain the role of profiles, implementation guides, and APIs.
- Assess whether FHIR-accessible data are complete and fit for a specific use case.
- Produce a FHIR data needs worksheet for an AI-supported workflow.

Module Overview

FHIR is increasingly important for public health because it supports structured, API-based access to health data. Public health staff do not need to become FHIR developers to participate in AI planning, but they do need to understand what FHIR can and cannot provide. This module explains how FHIR resources, profiles, implementation guides, and APIs may support AI-enabled public health workflows.

Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

FHIR matters for AI because it may make clinical and public health data more accessible, modular, and reusable. A FHIR API can allow an approved application to request specific resources, such as Patient, Observation, Condition, DiagnosticReport, Immunization, Encounter, Organization, or Task. This can support case summaries, decision support, cohort identification, immunization review, referral workflows, or public health reporting.

However, FHIR is not magic. A system may support FHIR but expose only certain data elements, use local extensions, omit important context, or require complex authorization. A vendor may say the product is FHIR-compatible while supporting only a narrow subset of resources. Public health staff need to ask whether the actual data required for the workflow are available, coded correctly, timely, and authorized for use.

FHIR use also raises governance questions. API access can make data retrieval easier, but easier retrieval does not remove privacy, minimum necessary, consent, legal authority, or public health reporting considerations. AI tools using FHIR data should have clear intended use, source traceability, access control, and validation.

Core Concepts

FHIR resource. A FHIR resource is a structured representation of a type of health information, such as Patient, Observation, Condition, Immunization, Encounter, DiagnosticReport, ServiceRequest, Organization, or Task. AI tools may use combinations of resources to support a workflow.

FHIR profile. A FHIR profile constrains or extends a resource for a specific purpose. Profiles help ensure that systems use resources consistently enough for implementation.

FHIR API. A FHIR API allows approved systems or applications to request, send, or update FHIR resources. API access should be governed through authentication, authorization, scopes, logging, and role-based access.

US Core. US Core defines profiles for core U.S. clinical data exchange. It can support common data access patterns, but public health workflows may require additional public health-specific profiles or implementation guides.

Implementation guide. A FHIR implementation guide defines how FHIR should be used for a particular use case. Public health IGs help translate general resources into operational exchange requirements.

Public Health Example

A health department may want an AI tool to summarize relevant EHR information for a case investigation. The tool might need Patient, Encounter, Condition, Observation, DiagnosticReport, MedicationRequest, and Practitioner or Organization resources. Staff need to verify whether the source system can provide these resources, whether the data include required dates and terminology, and whether the AI output cites the source data used.

An immunization use case may need Patient and Immunization resources, along with vaccine codes, dates, lot information, and dose context. If the FHIR source lacks historical immunizations or has incomplete patient matching, the AI tool may incorrectly identify gaps. The FHIR workflow must therefore be assessed for completeness, not just technical availability.

Technical Deep Dive

FHIR planning should begin with the use case rather than the technology. Staff should identify the public health decision or task, the data elements required, the FHIR resources that may contain those elements, the systems that expose those resources, and the authorization pathway. This prevents over-general claims about FHIR from substituting for a concrete data needs analysis.

Profiles and implementation guides matter because general resources can be used in different ways. A public health workflow may need specific required elements, value sets, extensions, or constraints. A FHIR resource that is technically valid may still lack a field required for surveillance, case investigation, or decision support.

Validation should test realistic workflows. Staff should verify not only that an API returns data, but that the returned data are complete, current, correctly coded, and usable in the AI-supported task. Testing should include missing data, conflicting data, multiple encounters, historical records, and user review of source traceability.

Risks, Failure Modes, and Guardrails

FHIR-compatible but incomplete. A product may support FHIR but not the data needed for the public health use case. Guardrails include data needs worksheets, test queries, and acceptance criteria.

Overbroad API access. AI tools may request more data than needed. Guardrails include scoped access, minimum necessary review, and audit logs.

Source traceability gaps. AI outputs may summarize FHIR data without showing which resources supported the output. Guardrails include source citations, resource identifiers, and user review.

Write-back risk. FHIR APIs may support updates or task creation. Guardrails include human approval, restricted permissions, sandbox testing, and rollback procedures.

Application to AI-Supported Workflows

Generative AI can use FHIR data to draft case or patient summaries, but the output should be grounded in visible source resources. Predictive analytics may use features derived from FHIR resources, which requires stable extraction and monitoring. RAG may combine FHIR-derived facts with agency guidance. Agentic workflows may create Tasks or update statuses, making permissions and approval steps essential.

Reflection Questions

Which public health workflows could benefit from structured FHIR data?

Which data elements are required for the workflow?

Are those elements available through FHIR in the source systems?

Which profiles or implementation guides apply?

How will staff verify that the AI tool used the correct FHIR data?

Practical Exercise

Create a FHIR data needs worksheet for one AI-supported public health use case. Include the use case, required data elements, relevant FHIR resources, source systems, API method, terminology requirements, validation questions, human review point, and documentation requirements.

Expected Artifact or Evidence

FHIR data needs worksheet

List of relevant FHIR resources

Vendor or implementation questions

Validation checklist

Expected Assignment

- FHIR data needs worksheet
- List of relevant FHIR resources
- Vendor or implementation questions
- Validation checklist

Knowledge Check

- 1. What is a FHIR resource?
- 2. Why does FHIR matter for AI-supported public health workflows?
- 3. What should staff verify when a vendor says a product is FHIR-compatible?
- 4. Which FHIR resource is commonly relevant to lab observations?
- 5. What is strong completion evidence for this module?

References and Resources

- HL7 FHIR resources: https://hl7.org/fhir/
- HL7 US Core Implementation Guide: https://build.fhir.org/ig/HL7/US-Core/
- HL7 eCR FHIR Implementation Guide: https://hl7.org/fhir/us/ecr/
- US Public Health Profiles Library
- ONC USCDI resources: https://isp.healthit.gov/united-states-core-data-interoperability-uscdi